MID: Theory of Computation, Date: 30 October 2022

A) Describe about **Chomsky hierarchy** (occasionally referred to as the **Chomsky–Schützenberger hierarchy**).

This hierarchy of grammars was described by Noam Chomsky in 1956 It is also named **after Marcel-Paul Schützenberger**, who played a crucial role in the development of the theory of formal languages.

Chomsky Hierarchy represents the class of languages that are accepted by the different machine. The category of language in Chomsky's Hierarchy is as given below:

1. Type 0 known as Unrestricted Grammar or Recursively enumerable grammar.
2. Type 1 known as Context Sensitive Grammar.
3. Type 2 known as Context Free Grammar.
4. Type 3 Regular Grammar

Describe all type.

B) Let, $\sum = \{0, 1\}$

Develop a DFA and make a transition table for the following language:

L = {w | w is a bit string which contains the substring 11}

C) Let, $\sum = \{0, 1\}$

A DFA for the following language:

L = {w | w is a binary string that has even number of 1s and even number of 0s}

Let the string 110101. Calculate the extension of transition to paths.

D) For the keyword recognizer occurrences of the words "include", Make a transition diagram.

E) Make a NFA:

Strings where the first symbol is present somewhere later on at least once somewhere later on at least once.

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